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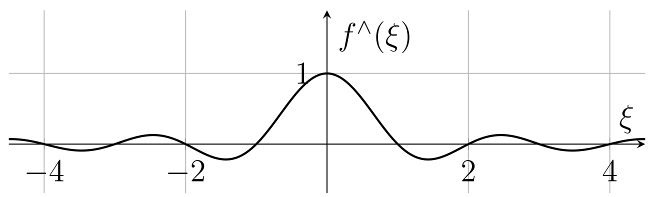
Ejem transformada fourier indicatriz intervalo

Ejemplo 1. Consideramos la función dada por $f(x) = \mathbb{1}_{[-1/2,1/2]}(x)$, entonces

$$\forall \xi \in \mathbb{R} \setminus \{0\} : f^\wedge(\xi) = \int_{-1/2}^{1/2} e^{-2\pi i x \xi}\,\mathrm{d}x = \left[\frac{e^{-2\pi i x \xi}}{-2\pi i \xi}\right]_{-1/2}^{1/2} = \frac{\sin(\pi\xi)}{\pi\xi}$$

y para $\xi = 0$, $f^\wedge(0) = \int_{-1/2}^{1/2} 1\,\mathrm{d}x = 1$.



Entonces, pese a que $f \in \mathcal{L}^1(\mathbb{R})$ y es de soporte compacto ($f \in \mathcal{C}_c(\mathbb{R})$), su transformada de Fourier $f^\wedge$ no tiene soporte compacto.

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